

BCA 5th Semester

JAVA

Topic : Important Java Programs

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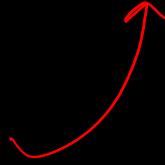
#	Program	Main Concept
1	Positive / Negative / Zero	if-else
2	Even / Odd	if-else
3	Greatest of 3 Numbers	Nested if
4	Grade Calculator	else-if
5	Leap Year	Conditions + operators
6	Multiplication Table	for
7	Sum of N Numbers	for
8	Factorial	for
9	Count Digits	while
10	Reverse Number	while

11	Palindrome Number	while + condition
12	Sum of Digits	while
13	Prime Number	Loop + condition
14	Armstrong Number	Loop + condition
15	Fibonacci Series	Loop
16	Factors of a Number	Loop + condition
17	Number Guessing Game	Loop + condition
18	ATM Menu	switch + loop

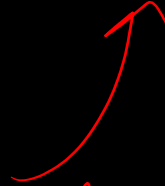
Positive / Negative / Zero



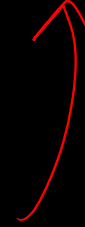
if



else-if



else



Even / Odd

2

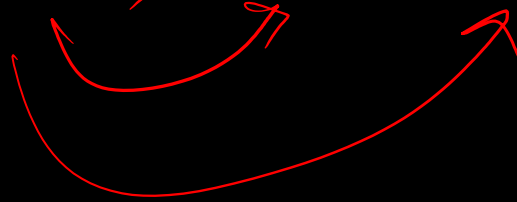
if - else

$$\frac{2}{2} = 0$$

Greatest of 3 Numbers

3

a, b, c



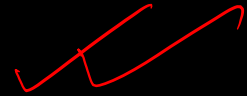
a

b > c

Grade Calculator

h

if/else - is - - - / else



H.W

h

~~Subject~~ → input

total → marks

Grade?

5

Leap Year

✓

→ 092

~~FF~~ ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) ✓

←

✓

~~FF~~

→ T

✓

Multiplication Table

6

n



10

5



loop

Sum of N Numbers

1

rules

$n = 5$

Sum = 0

Sum = Sum + n

0 + 1 = 1

1 + 2 = 3

3 + 3 = 6

6 + 4 = 10

10 + 5 = 15

Factorial

②

$$n = 5$$

$$5 \times 4 \times 3 \times 2 \times 1$$

$$n = 3$$

$$\text{fact} = 1$$

$$1 \times 3 = 3$$

$$3 \times 2 =$$

$$3 \times 3 =$$

9

Count Digits

253 ✓

```
int count = 0;

if (n == 0) {
    count = 1;
} else {
    n = Math.abs(n); ✓

    while (n > 0) {
        n = n / 10;
        count++;
    }
}
```

→ perfect ✓

→ $\left\{ \begin{array}{l} 253 / 10 = 25 \\ 25 / 10 = 2 \\ 2 / 10 = 0 \end{array} \right.$

$\left. \begin{array}{l} 1 \\ 2 \\ 3 \\ 1 \end{array} \right\}$

Reverse Number

253

10

```
int reverse = 0;
while (n != 0) {
    int digit = n % 10;
    reverse = reverse * 10 + digit;
    n = n / 10;
}
```

253 / 25 / 2 / 0 x

3 / 5 / 2

3 / 35 / 352

25 / 2 / 0

✓



Palindrome Number

```
int original = n; ✓✓
int reverse = 0;

while (n != 0) {
    int digit = n % 10;
    reverse = reverse * 10 + digit;
    n = n / 10;
}

if (original == reverse) {
    System.out.println(x: "Palindrome number");
} else {
    System.out.println(x: "Not a palindrome number");
}
```

121 = 1 2 1



Sum of Digits

12

```
n = Math.abs(n);  
  
int sum = 0;  
  
while (n != 0){  
    int digit = n % 10;  
    sum = sum + digit;  
    n = n / 10;  
}
```

positive

253

3 1 5 1 2
3 1 8 1 10
25 / 210

n = 0 / n > 0

Prime Number

13

i / n

$n/2$

loop

$n \% i$

25

6

1, 2, 3,

✓

```
boolean prime = true;
```

```
if (n <= 1) {  
    prime = false;  
} else {
```

```
    for (int i = 2; i < n; i++) {
```

```
        if (n % i == 0) {✓  
            prime = false;  
            break;  
        }
```

```
}
```


14

Armstrong Number

$$153 = 1^3 + 5^3 + 3^3 = 153$$

$$370 = 3^3 + 7^3 + 0^3 = 370$$

$$1634 = 1^4 + 6^4 + 3^4 + 4^4 = 1634$$

H.W

all num

3

```
int original = n;
int sum = 0;

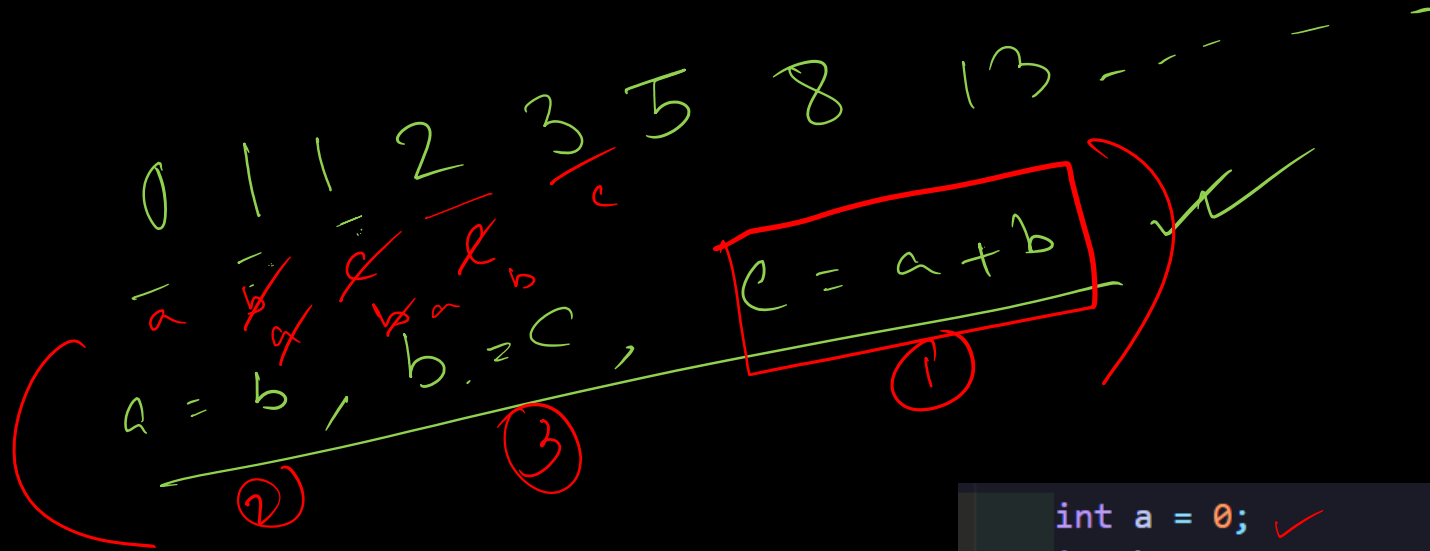
while (n != 0) {
    int digit = n % 10;
    sum = sum + (digit * digit * digit);
    n = n / 10;
}

if (sum == original) {
    System.out.println(x: "Armstrong number");
} else {
    System.out.println(x: "Not an Armstrong number");
}
```

Fibonacci Series

15

$a = 0$
 $b = 1$



```
int a = 0; ✓  
int b = 1; ✓  
  
System.out.print(s: "Fibonacci Series: ");  
  
for (int i = 1; i <= n; i++) {  
    System.out.print(a + " ");  
    int c = a + b;  
    a = b;  
    b = c;  
}
```

Factors of a Number

16

6 → 1, 2, 3, 6 ✓✓

10 → 1, 2, 5, 10

6

1 2 3 6

You, 20 hours ago | 1 author (You)

```
public class Factors {
```

Run | Debug

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        System.out.print(s: "Enter a number: ");
        int n = sc.nextInt();
```

```
        System.out.println(x: "Factors are:");
```

```
        for (int i = 1; i <= n; i++) {
```

```
            if (n % i == 0) {
                System.out.println(i);
            }
```

```
        sc.close();
```

Number Guessing Game

17

$n = 50$

60

50

40

ATM Menu

18